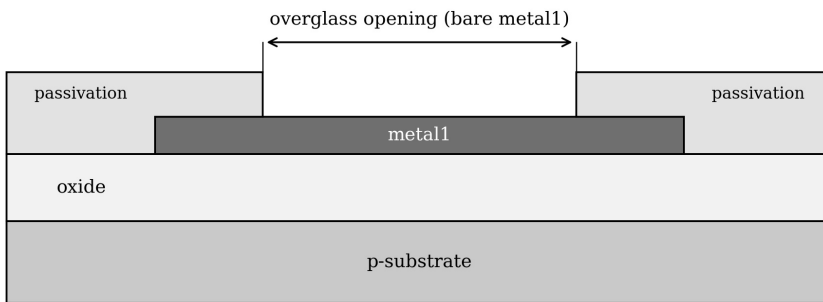
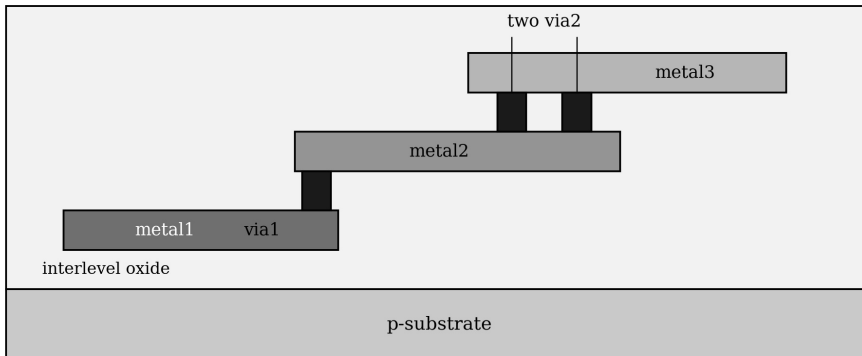


1.

It is a bonding pad. The overglass box is an opening in the passivation, not added material.



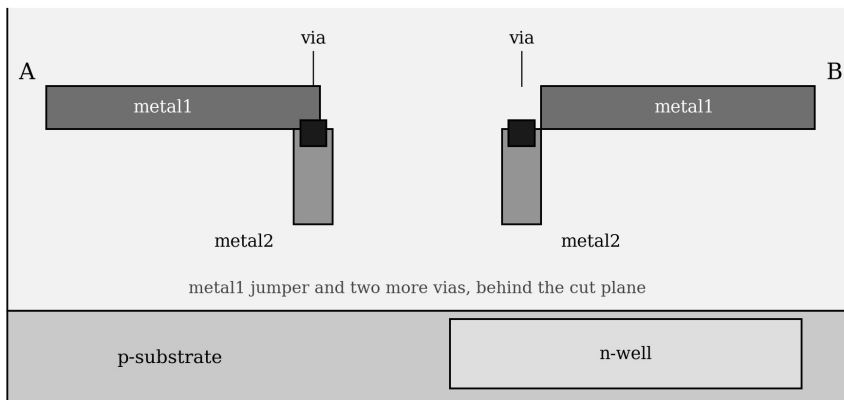
2.



Two via2 in parallel, 10 ohms each:

$$R = 10 / 2 = 5 \text{ ohms}$$

3.



Four vias in series. Using the 1.5 via as the scale, the metal is 2.5 wide and the path is 27.6 squares.

$$\text{metal} = 27.6 \times 0.1 = 2.8 \text{ ohms}$$

$$\text{vias} = 4 \times 10 = 40 \text{ ohms}$$

$$R(\text{A to B}) = 40 + 2.8 = 43 \text{ ohms}$$

The n-well carries no current, so it does not affect the resistance.

4.

$$\text{squares} = 10 \mu\text{m} / 100 \text{ nm} = 100, \text{ so } R = 100 \times 0.1 = 10 \text{ ohms}$$

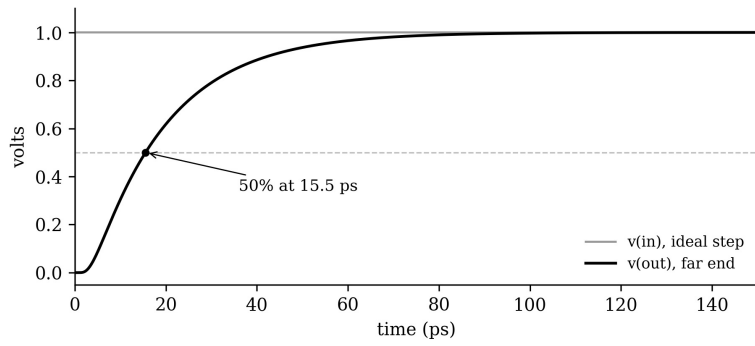
$$\text{loads} = 10 \mu\text{m} / 250 \text{ nm} = 40, \text{ so } C = 40 \times 100 \text{ fF} = 4 \text{ pF}$$

$$RC = 10 \times 4 \text{ pF} = 40 \text{ ps}$$

$$t(50\%) = 0.35 RC = 14 \text{ ps}$$

$$t(10 \text{ to } 90\%) = 0.9 RC = 36 \text{ ps}$$

Simulated with 40 sections of 0.25 ohm and 100 fF: 15.5 ps and 37.0 ps.



5.

40 pins is 10 pads per side, so 12 slots per side with the corners empty.

pad cell = $1500 / 12 = 125 \mu\text{m}$

gap = $125 - 75 = 50 \mu\text{m}$

centers = $62.5 + 125k \mu\text{m}$, for $k = 1$ to 10

drawn at 50 nm: pad 1500, cell 2500, die 30,000

